SOLUTION

SECTION–I

1. Ans. (A)

$\log(\sin^2 x.\cos x) = t$

$\Rightarrow (2\sin x.\cos^2 x - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x(1-\sin^2 x) - \sin^3 x)dx = dt.\sin^2 x.\cos x$

$\Rightarrow (2\sin x - 3\sin^3 x)dx = dt\,\frac{\sin x.\sin 2x}{2}$

$\Rightarrow \frac{(2-3\sin^2 x)}{\sin 2x}dx = \frac{dt}{2}$

$I = \frac{1}{2}\int tdt = \frac{t^2}{4} + C$

$= \frac{\log^2(\sin^2 x.\cos x)}{4} + C$

$= \log^2\left(\frac{1}{2}\sin 2x.\sin x\right) + C$

2. Ans. (1)

$\int \ell n\left(\sqrt{2+x} + \sqrt{2-x}\right)dx$ (By parts)

$= x\ell n\left(\sqrt{2+x} + \sqrt{2-x}\right),$

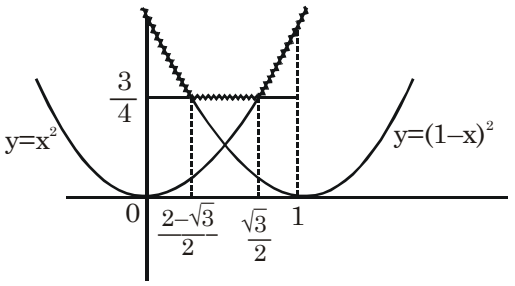
$-\int x.\frac{\left(\frac{1}{2\sqrt{2+x}} - \frac{1}{2\sqrt{2-x}}\right)}{\sqrt{2+x} + \sqrt{2-x}}dx$

$= x\log\left(\sqrt{2+x} + \sqrt{2-x}\right) - \frac{1}{4}\int \frac{4 - 2\sqrt{4-x^2}}{\sqrt{4-x^2}}dx$

$= xf(x) + \sin^{-1}\frac{x}{2} - \frac{x}{2} + C$

3. Ans. (A)

Sol.



$\int_0^1 f(x) = \int_0^{\frac{2-\sqrt{3}}{2}} (1-x)^2dx + \int_{\frac{2-\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{3}{4}dx + \int_{\frac{\sqrt{3}}{2}}^1 x^2dx$

$= \frac{6\sqrt{3}-1}{12}$

4. Ans. (A)

Sol. $I = \int_{\pi/3}^{\pi/6} \ell n\sin x dx + \int_{\frac{1}{2}\ell n\frac{3}{4}}^{-\ell n2} \sin^{-1} e^x dx$

$= -\frac{\pi}{6}\ell n2 - \frac{\pi}{6}\ell n\frac{3}{4} = \frac{\pi}{6}\ell n\frac{2}{3}$

5. Ans. (A)

Sol. $\int_0^y e^t dt + \int_0^x f(t)dt = g(x)$ .....(1)

diff. w.r.t. x

$e^y\frac{dy}{dx} + f(x) = g'(x)$

$\Rightarrow \frac{dy}{dx} = \frac{g'(x) - f(x)}{e^y}$ .....(2)

Again (1)

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$$\Rightarrow \quad e^y - 1 = g(x) - \int_0^x f(t)dt$$

$$\Rightarrow \quad e^y = 1 + g(x) - \int_0^x f(t)dt$$

$$\therefore \quad (2) \quad \Rightarrow \quad \frac{dy}{dx} = \frac{g'(x) - f(x)}{1 + g(x) - \int_0^x f(t)dt}$$

6. Ans. (A)

$$\int e^x \left(\frac{1}{\ell nx} - \frac{1}{(x)(\ell nx)^2} \right) dx$$

$$\int \frac{e^x}{\ell nx} dx - \int \frac{e^x}{x(\ell nx)^2} dx$$

$$\left[\frac{1}{\ell nx} e^x - \int \frac{-1}{(\ell nx)^2 x} e^x dx \right] - \int \frac{e^x}{x(\ell nx)^2} dx + C$$

(Applying integration by parts)

$$= \frac{e^x}{\ell nx} + C$$

7. Ans. (A)

$$\lim_{x \to 0} \frac{\ell n(1+x) + \frac{x}{1+x}}{f(x).\sin x} = 5$$

$$\lim_{x \to 0} \frac{\frac{\ell n(1+x)}{x} + \frac{1}{1+x}}{f(x).\frac{\sin x}{x}} = 5$$

$$\frac{2}{5} = f(0)$$

8. Ans. (A)

$$I_2 = \int_0^1 \frac{x^2}{e^{x^3}(2-x^3)} dx = \frac{1}{3} \int_0^1 \frac{dz}{e^{1-z}(z+1)}, (z = 1 - x^3)$$

$$= \frac{1}{3} \int_0^1 \frac{e^z}{e(z+1)} dz = \frac{1}{3e} I_1$$

$$\frac{I_1}{I_2} = 3e$$

9. Ans. (A,C,D)

$$\int \frac{\sec^2 x + 3}{\sqrt{\sec^2 x + 3}} dx = \int \frac{\sec^2 x dx}{\sqrt{4 + \tan^2 x}} + \int \frac{3\cos x}{\sqrt{4 - 3\sin^2 x}} dx$$

$$= \int \frac{dt}{\sqrt{4 + t^2}} + \int \frac{\sqrt{3}\, d\theta}{\sqrt{4 - \theta^2}}$$

$$= \ell n \left| t + \sqrt{4 + t^2} \right| + \sqrt{3} \sin^{-1} \left(\frac{\theta}{2} \right) + C$$

$$= \ell n \left| \tan x + \sqrt{4 + \tan^2 x} \right| + \sqrt{3} \sin^{-1} \left(\frac{\sqrt{3} \sin x}{2} \right) + C$$

$f(x) = \tan x$, $g(x) = \sin x$, $k = \sqrt{3}$

10. Ans. (A,B,C,D)

Let $I = \int (2x^6 + 15x^4 + 2x^2 + 3)\cos 2x\, dx$

$= \int (2x^6 + 2x^2).\cos 2x + \int (15x^4 + 3).\cos 2x dx$

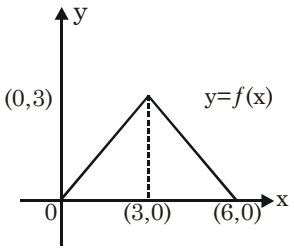
$\therefore$ Apply integrate by part, we get

$= (x^6 + x^2 + 1)\sin(2x) + (3x^5 + x).\cos(2x) + k$

$\therefore \quad f(x) = (x^6 + x^2 + 1)$; $g(x) = (3x^5 + x)$.

11. Ans. (A,C)

$f(x)$ is periodic with period '6'



$$\therefore \int_0^6 f(x)dx = \frac{1}{2} \times 6 \times 3 = 9$$

12. Ans. (A,C)

$$\int_0^x f(t)dt + \int_1^{f(x)} g(t)dt = f'(x)$$

Using : $\int_a^b f(x)dx + \int_{c=f(a)}^{d=f(b)} f^{-1}(x)dx = bd - ac$

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$\Rightarrow x.f(x)-0\times1=f'(x)$ (think !)

$\Rightarrow \dfrac{f'(x)}{f(x)}=x \qquad (\because f(x)\neq 0)$

$\Rightarrow \ell n f(x)=\dfrac{x^2}{2}+C$

put $x=0 \Rightarrow C=0$

$\Rightarrow f(x)=e^{\frac{x^2}{2}}$

$f(\sqrt{2})=e \Rightarrow g(e)=\sqrt{2}$

$f(1)=\sqrt{e}$

13. Ans. (B)

(P) $\lim\limits_{n\to\infty}\int\limits_0^2\left(1+\dfrac{t}{n+1}\right)^n dt$

$$\frac{\left[\left(1+\frac{t}{n+1}\right)^{n+1}\right]_0^2}{(n+1)\left(\frac{1}{n+1}\right)}$$

$\lim\limits_{n\to\infty}\left(1+\dfrac{2}{n+1}\right)^{n+1}-1=\boxed{e^2-1}$ $\boxed{P\to3}$

(Q) $f'(x)=f(x)$, $f(0)=1$

$f(x)+g(x)=x^2$

$f(x)=e^x$

$g(x)=x^2-e^x$

$\int\limits_0^1 e^x(x^2-e^x)\,dx$

$\int\limits_0^1 (x^2e^x-e^{2x})\,dx$

$\left[x^2e^x-2\int xe^x-\dfrac{e^{2x}}{2}\right]_0^1$

$\left[x^2e^x-2(xe^4-e^x)-\dfrac{e^{2x}}{2}\right]_0^1$

$\left(e-2(e-e)-\dfrac{e^2}{2}\right)-\left(2-\dfrac{1}{2}\right)$

$e-\dfrac{e^2}{2}-\dfrac{3}{2}$ $\boxed{Q\to1}$

(R) $\int\limits_0^1 e^{e^x}(1+xe^x)\,dx$

$e^x=t \Rightarrow e^x dx=dt \Rightarrow dx=\dfrac{dt}{t}$

$\int\limits_1^e e^t(1+\ell nt.t)\dfrac{dt}{t}$

$\int\limits_1^e e^t\left(\ell nt+\dfrac{1}{t}\right)dt$

$\left[e^t\ \ell nt\right]_1^e \Rightarrow \boxed{e^e-0}$ $\boxed{R\to4}$

(S) $\lim\limits_{K\to0}\dfrac{1}{K}\int\limits_0^K(1+\sin 2x)^{\frac{1}{x}}\,dx$

$\lim\limits_{K\to0}\left((1+\sin 2K)^{\frac{1}{K}}\right)$

$e^{\lim\limits_{K\to0}\left(\frac{\sin 2K}{2K}\right)^2}=e^2$ $\boxed{S\to2}$

SECTION–III

1. Ans. 2

Sol. $\alpha=\int\limits_{2017}^{2019}(x-2017)^2 f(x-2017)\,dx$

$x-2017=t$

$\alpha=\int\limits_0^2 t^2 f(t)\,dt$

$\beta=\int\limits_{-2019}^{-2015}(x+2017)^2 f(x+2017)\,dx$

$x+2017=z$

$\beta=\int\limits_{-2}^{2} z^2 f(z)\,dz=2\alpha$

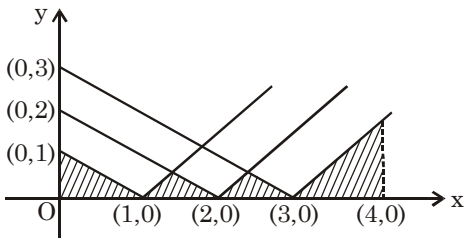
$\dfrac{\beta}{\alpha}=2$

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2. Ans. 5

Sol.



Required value

$$=\frac{1}{2}(1)(1)+2\left(\frac{1}{2}\times1\times\frac{1}{2}\right)+\frac{1}{2}\times1\times1=\frac{3}{2}$$

3. Ans. 9

4. Ans. 9

Sol. $f(x)=-\int_0^1 f^2(t)dt-2x\int_0^1 f(t)dt$

$\therefore \quad f'(x)=-2\int_0^1 f(t)dt \qquad .....(i)$

$=$ constant

$\Rightarrow$ f must be linear.

So, let $f(x) = (ax + b)$

$\therefore \quad a=-2\int_0^1 (at+b)dt$

& $\quad b=-\int_0^1 (at+b)^2dt$

$\Rightarrow$ $a = 3, b = -3$

$\Rightarrow$ $f(x) = 3(x - 1)$

So, $f(4) = 3(4 - 1) = 9$

5. Ans. 2

Sol. $\frac{1}{n\pi}.2n\int_0^1 \frac{dx}{\sqrt{\{x\}\{-x\}}}=\frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{x(1-x)}}$

$$=\frac{4}{\pi}\left(\sin^{-1}\sqrt{x}\right)_0^1=2$$

6. Ans. 7

Sol. $I_7=\int_0^1 \underset{I}{\left(1-x^4\right)}.\underset{II}{1}\,dx$

$I_7=\left[x\left(1-x^4\right)^7\right]_0^1-\int_0^1 7\left(1-x^4\right)^6\left(-4x^3\right).xdx$

$I_7=0+28\int_0^1 \left(1-x^4\right)^6x^4dx$

$I_7=28\int_0^1 \left(1-x^4\right)^6\left[1-\left(1-x^4\right)\right]dx$

$I_7=28I_6-28I_7 \Rightarrow 29I_7=28I_6$

$\Rightarrow \frac{29I_7}{4I_6}=7$

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